

# Shai Aviram Bergman

Principal Researcher • Team & Technical Lead • Huawei Zurich Research Center

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## PROFILE

Systems researcher and team lead working across operating systems, virtualization, memory management, and storage — and increasingly at the intersection of systems and AI, where principled systems design makes AI faster, more efficient, and more capable. I lead a research team at Huawei's Zurich Research Center on storage systems that accelerate LLM inference, CXL-based memory, and storage architectures for modern data centers; I also contributed to the specification of Huawei's UnifiedBus (UB) interconnect. Ph.D. from the Technion; author of peer-reviewed papers at top systems and architecture venues (USENIX ATC, EuroSys, ISCA, HPCA, Middleware, ISMM, MSST), with a Best Paper Award at USENIX ATC 2025; inventor on five published patent families — with patents granted in the US and Europe — and further applications pending.

## EXPERIENCE

### Principal Researcher • Team & Technical Lead

May 2025 – present

#### Huawei Zurich Research Center

Zurich, Switzerland

- Lead a research team working at the intersection of storage and AI: storage systems that accelerate LLM inference and agentic-AI workloads, alongside CXL-based memory systems for modern data centers.
- **KV-cache systems for LLM inference.** Persist KV caches to storage and reuse them beyond exact prefix matches via selective recomputation of a small token fraction; offload caches across GPU memory, DRAM, and NVMe; manage storage bandwidth to sustain cache loading under concurrent inference — prototyped on vLLM, SGLang, and LMCahe.
- **RAG serving.** Speculative decoding that drafts tokens from the retrieval datastore and verifies them in parallel on the target model, removing the separate draft model; approximate caching of retrievals across similar queries.
- **Systems for agentic AI.** New system interfaces and file systems designed for AI agents, including offloading agents' data operations into storage (in-storage processing).

### Senior Researcher

Aug 2023 – May 2025

#### Huawei Zurich Research Center

Zurich, Switzerland

- **Distributed caching over memory fabrics.** Designed an OS-level distributed page cache for CXL 3.0 shared memory that keeps a single DRAM copy of each file page across nodes while preserving POSIX semantics, eliminating redundant caching and I/O — up to 12.4× speedups in multi-host emulation.
- **UnifiedBus.** Contributed to the specification of UnifiedBus (UB), Huawei's openly published interconnect for peer-to-peer resource pooling in SuperPoD-scale AI systems.
- **In-storage processing.** Researched computational storage, including accelerating approximate-nearest-neighbor vector search inside storage devices to cut data movement for retrieval and vector-database workloads.
- **MoE inference offload.** Enabled Mixture-of-Experts LLM inference on memory-constrained servers by offloading expert (FFN) weights to fast storage and streaming the experts activated per token.
- **Power-efficient storage servers.** Co-led the design of an I/O-metric-guided CPU power-management subsystem (DVFS and core power-gating) for all-flash storage servers, improving IOPS per watt on production traces and saving power on MLPerf Storage workloads at no performance cost.
- **SMT scheduling.** Devised symbiosis-aware scheduling that co-locates threads with complementary resource profiles on SMT cores, raising throughput without hardware changes.

### Chief Architect, Advanced Technologies

2022 – 2023

#### UniFabrix

Haifa, Israel

- **CXL memory pooling.** Architected dynamic borrowing and lending of memory across hosts over CXL, built on Linux memory hot-plug; hardware/software co-design and OS support for disaggregated memory.
- **Performance frameworks.** Developed frameworks that let applications consume pooled memory without performance loss — and with gains where the pool's added bandwidth is exploited.

### Visiting Scholar

2022

#### Imperial College London

London, UK

- Designed a memory-virtualization mechanism that lets VMs manage flat guest-virtual-to-host-physical page tables, with inter-VM isolation enforced through per-VM physical-memory tagging repurposed from confidential-computing hardware; with Prof. Peter Pietzuch and Prof. Lluís Vilanova.
- Prototyped in Linux/KVM/QEMU on x86-64: near-native paging performance on 100-GiB-scale data-center workloads — up to 2.4× faster than nested paging (Intel EPT).

- Lecturer and teaching-assistant-in-charge for the “Structure of Operating Systems” course; mentored graduate students in operating-systems research projects and papers throughout their studies.
- **Nested virtualization.** Co-led research on accelerating nested virtualization by safely offloading the guest hypervisor’s hot-path exit handling into the bare-metal hypervisor as kernel-verified eBPF programs.
- Research in accelerated computing systems: accelerators, disaggregated memory, memory tiering, virtualization, and SSDs.

**Software Architect, Innovation Team**

2019 – 2021

## Toga Networks – a Huawei company

Haifa, Israel

- **Consultant (2020–2021).** Architected memory management that removes memory-pinning requirements for next-generation RDMA NICs: NIC-side on-the-fly address translation with page-fault handling and requester/responder memory-status signaling – three patent families (US and EP patents granted).
- Designed efficient intra-host data movement for UCX.
- **Intern (2019).** Linux kernel memory-management development for HPC systems.

**Software Engineer, Runtime Team**

2016

## LogicBlox

Berkeley, CA, USA

- Implemented and optimized vectorized (SIMD) query execution in the LogicBlox 4 runtime – a commercial Datalog (LogiQL) database engine – and investigated GPU offload for accelerating query evaluation.

**Co-Op Engineer, Innovation Team**

2015

## AMD

Tel Aviv, Israel

- Prototyped OS support for transparent peer-to-peer DMA between NVMe SSDs and GPUs with standard POSIX file semantics, removing the CPU from the data path – paving the way for later GPU-direct storage technologies such as NVIDIA GPUDirect Storage (GDS).

Earlier: Software Developer (VoIP / systems / web), Voicelinx, London (2010–2015). Staff Sergeant, Israel Defense Forces (2007–2010).

**EDUCATION****Ph.D., Electrical & Computer Engineering**

2017 – 2022

## Technion – Israel Institute of Technology

Thesis: “Memory Management for Emerging Memory Technologies in Future Data-Centers.” Advisor: Prof. Mark Silberstein.

- **Disaggregated memory.** Showed classic OS memory optimizations falter at far-memory latencies and built a Linux memory-tiering subsystem that maximizes local-memory hits, outperforming state-of-the-art tiered-memory systems (ISMM ’22).
- **Storage.** Co-designed the Linux swap subsystem with Zoned-Namespace SSDs – 10× lower 99th-percentile latency for memcached (USENIX ATC ’22; ACM TOS ’23).
- **Virtualization & hardware.** Devised near-native memory virtualization: guest-managed flat page tables with isolation via CPU physical-memory tagging (USENIX ATC ’23).

**M.Sc., Electrical & Computer Engineering**

2015 – 2017

## Technion – Israel Institute of Technology

Thesis: “High-Performance Disk I/O on GPUs” – OS-integrated peer-to-peer DMA between NVMe SSDs and GPUs behind standard POSIX file I/O (USENIX ATC ’17). *Cum Laude* (thesis 95%, courses 96.7%).

**B.Sc., Electrical & Computer Engineering**

2011 – 2015

## Technion – Israel Institute of Technology

Specialization in Computers (hardware & software) and Networks. *Cum Laude* (average 90.2%).

**PUBLICATIONS**

1. **GenPairX: A Hardware-Algorithm Co-Designed Accelerator for Paired-End Read Mapping.** J. Eudine, C. Li, Z. Cheng, R. Andri, C. Firtina, M. Sadrosadati, N. Mansouri Ghiasi, K. Koliogeorgi, A. Nag, A. Tavakkol, H. Mao, O. Mutlu, **S. Bergman**, J. Zhang. *IEEE Int. Symp. on High-Performance Computer Architecture (HPCA)*, 2026.
2. **DPC: A Distributed Page Cache over CXL.** **S. Bergman**, Z. Yang, J. Eudine, G. Negro, O. Mutlu, A. Tavakkol, J. Zhang. *arXiv:2604.19494*, 2026.
3. **Accelerating Nested Virtualization with HyperTurtle.** O. Ben Zur, J. Krebs, **S. Bergman**, M. Silberstein. *USENIX Annual Technical Conf. (ATC)*, 2025. **BEST PAPER AWARD**
4. **Leveraging Approximate Caching for Faster Retrieval-Augmented Generation.** **S. Bergman**, A.-M. Kermarrec, D. Petrescu, R. Pires, M. Randl, M. de Vos, J. Zhang. *ACM/IFIP Int. Middleware Conf. (Middleware)*, 2025.
5. **Storage Cluster for Persistency, CXL Pools for Caching.** K. Manaouil, J. Zhang, Z. Yang, Z. X. Wang, **S. Bergman**, A. Barbalace. *Workshop on Heterogeneous Composable and Disaggregated Systems (HCDS)*, 2025.
6. **Intent-Driven Storage Systems: From Low-Level Tuning to High-Level Understanding.** **S. Bergman**, W. W. Song, L. Cavigelli, K. Berestizshevsky, K. Zhou, J. Zhang. *arXiv:2510.15917*, 2025.
7. **Composable Storage Servers: A Storage Paradigm for Disaggregated Systems.** **S. Bergman**, O. Mutlu, W. Yong, K. Huang, J. Zhang. *IEEE Int. Conf. on Networking, Architecture and Storage (NAS)*, 2024.

8. **Answering the Call to ARMs with PACER: Power-Efficiency in Storage Servers.** H. Cao, [S. Bergman](#), S. Sun, Y. A. Zhou, X. Li, J. Gao, Z. Cheng, J. Zhang. *Int. Symp. on Mass Storage Systems and Technologies (MSST)*, 2024.
9. **Flexible Swapping for the Cloud.** M. Pandurov, L. Humbel, D. Sepp, A. Ttofari, L. Thomm, D. L. Quoc, S. Chandrasekaran, S. Santhanam, C. Ye, [S. Bergman](#), et al. *arXiv:2409.13327*, 2024.
10. **Reducing the Virtual Memory Overhead in Nested Virtualization.** O. Ben Zur, [S. Bergman](#), M. Silberstein. *ACM Int. Systems and Storage Conf. (SYSTOR)*, Poster, 2023.
11. **Translation Pass-Through for Near-Native Paging Performance in VMs.** [S. Bergman](#), M. Silberstein, T. Shinagawa, P. Pietzuch, L. Vilanova. *USENIX Annual Technical Conf. (ATC)*, 2023.
12. **ZNSwap: Un-Block Your Swap.** [S. Bergman](#), N. Cassel, M. Björling, M. Silberstein. *ACM Transactions on Storage (TOS)*, 2023.
13. **ZNSwap: un-Block your Swap.** [S. Bergman](#), N. Cassel, M. Björling, M. Silberstein. *USENIX Annual Technical Conf. (ATC)*, 2022.
14. **Reconsidering OS Memory Optimizations in the Presence of Disaggregated Memory.** [S. Bergman](#), P. Faldu, B. Grot, L. Vilanova, M. Silberstein. *ACM SIGPLAN Int. Symp. on Memory Management (ISMM)*, 2022.
15. **Slashing the Disaggregation Tax in Heterogeneous Data Centers with FractOS.** L. Vilanova, L. Maudlej, [S. Bergman](#), T. Miemietz, M. Hille, N. Asmussen, M. Roitzsch, H. Härtig, M. Silberstein. *European Conf. on Computer Systems (EuroSys)*, 2022.
16. **SPIN: Seamless Operating System Integration of Peer-to-Peer DMA Between SSDs and GPUs.** [S. Bergman](#), T. Brokhman, T. Cohen, M. Silberstein. *ACM Transactions on Computer Systems (TOCS)*, 2019.
17. **ActivePointers: A Case for Software Address Translation on GPUs.** S. Shahar, [S. Bergman](#), M. Silberstein. *ACM SIGOPS Operating Systems Review*, 2018.
18. **SPIN: Seamless Operating System Integration of Peer-to-Peer DMA Between SSDs and GPUs.** [S. Bergman](#), T. Brokhman, T. Cohen, M. Silberstein. *USENIX Annual Technical Conf. (ATC)*, 2017.
19. **ActivePointers: A Case for Software Address Translation on GPUs.** S. Shahar, [S. Bergman](#), M. Silberstein. *Int. Symp. on Computer Architecture (ISCA)*, 2016.

## PATENTS

1. **Unaware Fine-Grained VM Memory Pinning for Passthrough Devices.** A. Geron, B. Belkar, X. Mao, [S. Bergman](#). WO2026118061A1, *pending*, 2026. Guest-transparent, fine-grained pinning: a pass-through device asks the hypervisor to pin only the guest pages it uses for DMA, instead of all VM memory.
2. **Data Processing Unit.** [S. Bergman](#), B. Belkar, B. Ben Ishay, J. Zhang. WO2026057153A1, *pending*, 2026. A DPU that presents remote and pooled memory to the host as ordinary byte-addressable memory, offloading translation, caching, and page-fault handling over RDMA/CXL.
3. **Devices and Methods for Remote Direct Memory Access.** B. Belkar, D. Ganor, A. Geron, [S. Bergman](#). WO2022156902A1, *national phase pending*, 2022. RDMA protocol extension in which the responder signals memory-region status to the requester, making RDMA to unpinned memory reliable.
4. **Method and System for Direct Memory Access.** B. Belkar, A. Margolin, [S. Bergman](#), R. Hyatt, D. Volkind, L. Khernosh, T. Brokhman. WO2021239230A1; *granted as US 12,174,751 B2*, 2021. NIC-side address translation mapping DMA addresses onto physical memory or the swap/storage tier – no pinned regions required.
5. **Method and System for Direct Memory Access.** B. Belkar, A. Margolin, [S. Bergman](#), R. Hyatt, D. Volkind, L. Khernosh, T. Brokhman. WO2021239228A1; *granted as US 12,182,034 B2 & EP 4133375 B1*, 2021. Lets the NIC write RDMA data into non-pinned memory, redirecting writes to unmapped or swapped-out pages and updating translations on the fly.

Five additional patent applications pending (unpublished).

## AWARDS & HONORS

- Best Paper Award, USENIX ATC 2025 — *HyperTurtle*
- President's Award — Significant Business Contribution, Huawei, 2026
- Technology Innovation Award, Huawei, 2026
- European Research Institute Excellent Contributor, Huawei, 2025
- Future Star Award, Huawei (×2, 2024–2025)
- Technical Innovation Award, Huawei, 2024
- 1st Place, Mellanox BlueField Hackathon, 2019
- Outstanding Teaching Assistant, Technion (×5, 2016–2019)
- M.Sc. & B.Sc. *Cum Laude*, Technion
- Technion President's List, 2015

## PROFESSIONAL SERVICE

**Program committees:** HPCA 2027, EuroSys 2027, ASPLOS 2026, OSDI 2025, EuroSys 2024. **Teaching:** Lecturer, Structure of Operating Systems (046209), Technion, 2022–2023.

## SKILLS

**Systems.** Linux kernel development (mm, block, KVM, NVMe), virtualization (KVM/QEMU), CXL & disaggregated/tiered memory, RDMA & SmartNICs/DPUs, eBPF, GPU/accelerator systems, storage & SSDs (ZNS, SPDK)

**AI inference systems.** LLM serving (vLLM, SGLang, LMCACHE, KTransformers), KV-cache management & reuse, speculative decoding, RAG, MoE offloading, in-storage processing

**Languages & tools.** C, C++, CUDA, Python, Bash; Java, JavaScript

## LANGUAGES

- English — native
- Hebrew — native